

Connected Specimens

A low-cost framework for museums to expand the stories of displayed specimens

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ABSTRACT

The specimens nowadays on display in museums are the attempts and efforts of human wisdom to create traces of species that once existed, to represent their appearance, and, for the species that are still here, to resist their disappearance. But there is an experiential divide between the stories offered by museums and visitors' desire to go further, between digital interfaces and physical settings, and between developed and less developed regions. This project creates a low-cost, data-driven framework that leverages the scalability of digital media and the perceptibility of real-world objects to breathe new life into exhibits and enhance in-person engagement in museums. The project will first focus on bird species at the National Wetland Museum of China, enhancing visitors' cognitive experience of the specimens through a digital-physical integration strategy. The future vision of the project is to open up to other species datasets and provide APIs and SDKs for further development.

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INTRODUCTION

Species are disappearing from the world every hour, minute, second, of every day. “More than 35,500 species are threatened with extinction. That is still 28% of all assessed species,” IUCN¹ writes as the headline on their website. Making specimens of plants and animals and displaying them in museums are the attempts and efforts of mankind's wisdom in making authentic copies of the species on earth, creating traces of their presence, representing their appearance, and preserving this "disappearance".

However, in most museums, the static specimens that are displayed often provide limited information. They are either arranged like a matrix on the wall or housed in glass cases. Visitors have no way to get more information than what they look like, and even their looks are observed from a limited perspective. There is a gap between the stories museums provide and what people further desire.

The good news, however, is that there are already a number of practices in the realm of museum informatics to bridge this gap by expanding the story of specimens through digitization. For example, the Division of Birds, National Museum of Natural History (NMNH), Smithsonian Institution “maintains the third largest bird collection in the world and 85% of their total holdings are searchable through the public database. Users can study birds’ skins, skeletons, fluid preserved, eggs, nests, and mounted skins²” in this database. We also can see that plenty of museums have developed virtual field-trip experiences especially in the era of Covid-19 pandemic.

To sum up, many actions are being made by museums to digitize their on-site collections, either by photographing or scanning and reconstructing them into 3D assets, to open them to a wider public. These attempts are very valuable practices that focus on openness of data. However, the digital experience and museum exhibits are often isolating. This creates

¹ Established in 1964, The International Union for Conservation of Nature’s Red List of Threatened Species has evolved to become the world’s most comprehensive information source on the global conservation status of animal, fungi and plant species.

² Search the Division of Birds Collections. <https://collections.nmnh.si.edu/search/birds/>

a gap between the virtual world and the physical collections and does not build dialogues or conversations between them. If we can create digital galleries and ultimately combine them with museum exhibits to make them mutually complementary by taking full advantage of scalability of digital media and perceivability of real-world objects, we can revitalize or reactivate those exhibits and enhance the in-person engagement in museums.

Many museums are aware of the gap between the virtual world and the physical one, and have adopted some cutting-edge attempts. For example, NMNH created digital-to-physical conversations by setting up an kiosk-based Collection Browser that can lead visitors to find a specific specimen³ in Q?rius⁴, which is a science-based center where visitors can handle and interact with real artifacts, at the end of 2013. Not coincidentally, University of Michigan's Museum of Natural History unveiled an exhibition in 2018, "allows visitors to examine 25 real artifacts and use them to trigger interactive stories about the collection"⁵. They created physical-to-digital conversations by developing profiles for each specimen and created an RFID system to help visitors discover specimens at a physical level.

However, there is still a gap between the developed museums and the less developed ones. We cannot deny that the aforementioned cutting-edge practices have brought new insights into museum curation, but the installations they use are often highly customized, such as Kiosk, multi-touch tables, interaction projection, costly and with specificity; but the realities we must face are:

1. Not all museums have sufficient funds and resources to upgrade and renovate their infrastructure and hire a professional agency to customize a gorgeous interactive interface, especially in less developed regions.

³ "Q?Rius", Potiondesign.Com, 2013, <https://www.potiondesign.com/project/qrius/>.

⁴ Lonnae Parker, "Smithsonian's Q?Rius Gallery Invites Teens To Explore Science", *The Washington Post*, 2013, https://www.washingtonpost.com/lifestyle/style/smithsonians-qrius-gallery-invites-teens-to-explore-science/2013/12/12/7a1d21b4-635e-11e3-aa81-e1dab1360323_story.html.

⁵ "Natural History Exhibit Allows Visitors To Examine Real Artifacts", 2018, <https://ideum.com/portfolio/rfid-table>.

2. Not all museums' collections are rich enough to allow visitors to access and handle directly, even the University of Michigan's Museum of Natural History, I mentioned above, has customized unique stories for only 25 specimens.

In this project, I proposed solutions to progressively bridge the three gaps above:

1. A data-driven approach to connect isolating avian specimens displayed in the museum, uncover the overlapping geographical habitats, reveal the potential relationships in taxonomy and genealogy, map the links between threats and conservation actions we can take, and thus bridges the gap between the stories museums provide and people further desire.
2. A two-way conversation experience to expand the stories of the displayed specimens. To expand the stories of the displayed specimens in museums by creating digital exploratory interfaces without losing the link of physical settings. That means, you can search objects in your device by recognizing physical specimens and you can build connections as guides between digital specimens and physical ones to identify them in the real world settings.
3. An available, accessible and affordable add-on digital framework that provides a universal workflow to support museums in different developed regions or at different levels of development to achieve a high level of display at a low cost, without the modification of existing infrastructure.

At this stage, I tentatively focus on avian species in the National Wetland Museum of China (Xixi Museum). In the future, this work could be extended as an universal platform in terms of the species datasets, the adaptation for different museums and the implementation process.

Considering that our data is from IUCN Red Lists, the most comprehensive information source on the global conservation status of species, the data structures of different species are identical, so this application could be compatible with datasets of other species, such as

mammals, amphibians, fishes and trees, as well. In addition, the species exhibited in different museums can be seen as a subset of this largest database, so we can tailor the subset in the application for each museum according to their corresponding species. Finally, this application is delivered with basic features to maintain its universal properties and an API will also be provided for further in-depth development.

SYSTEM DESIGN

Overview

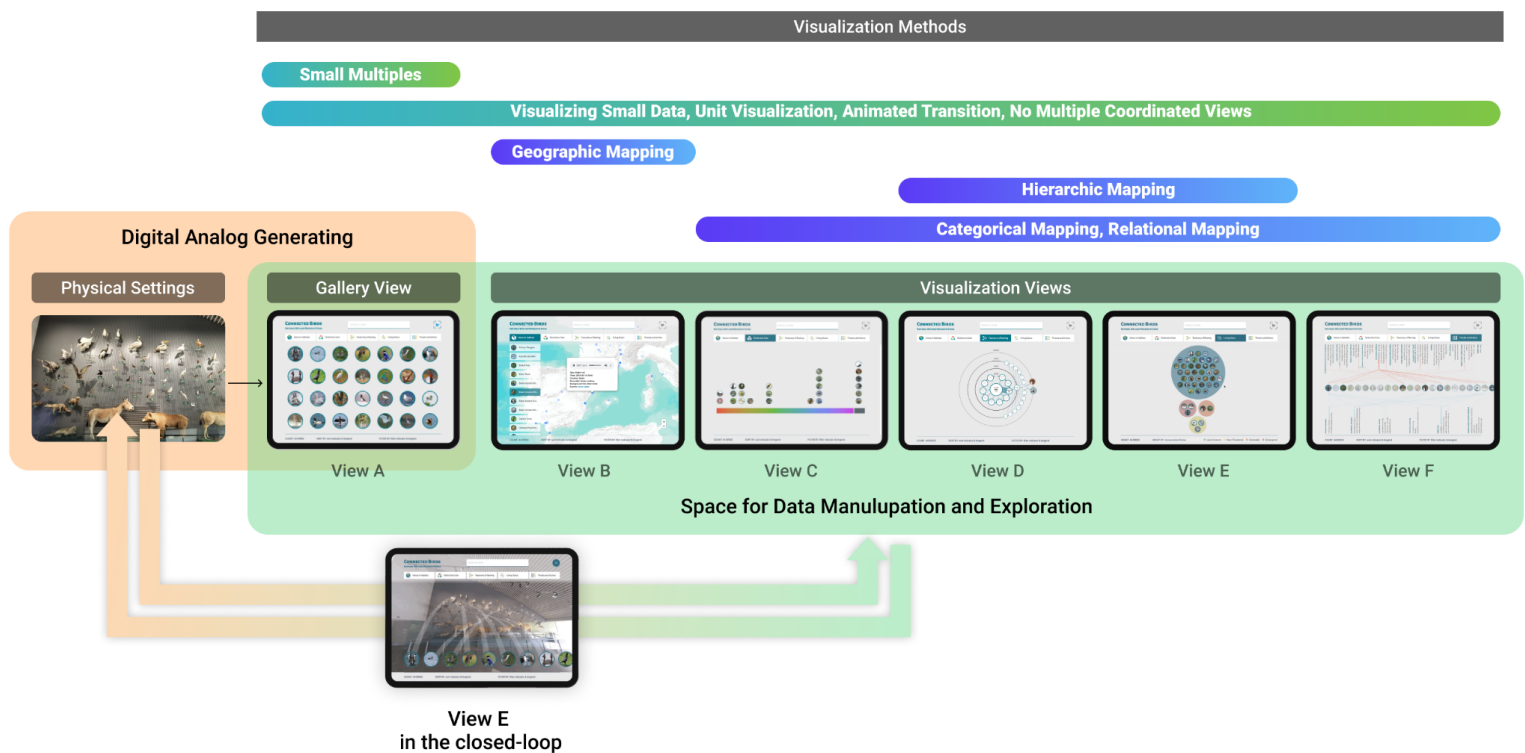


Figure 1: The structure of the application and the logically closed-loop among views.

The application⁶ consists of a gallery view and multiple thematic views. The Gallery View (View A in Figure 1) is an analog that includes 44 digital replicas, or mirror objects, of the real-setting specimens in Xixi Museum (this process is shown in the orange area in Figure 1). The digital analog creates an extended space (the green area in Figure 1) for data manipulation and exploration, which provides an opportunity for users to reorganize, re-sort,

⁶ Access the application: <https://connected-specimens.com>

and apply interactions to those static specimens, forming the augmentations towards the real objects. This concept could be more understandable by analogy with sandbox — a testing environment where unknown software can be run but will not affect the production environment. The digital extended space here is like a sandbox where the virtual mirror specimens could be further explored but with no side effects to the real ones.

In this extended space, a variety of information visualizations — geographic, hierarchic, relational, and categorical — are introduced to support visual exploration, which underlie the five modes (View B to View F in Figure 1) of engagement with a theme — through bird sound, geographic distribution, feather color, species taxonomy, conservation status, population trend, and threats and actions, etc. A visualization method may support many themes, and in turn, one theme may be supported by many visualization methods (this many-to-many relationship is shown by the gantt-chart-like lines in Figure 1).

Finally, the recognition component (View E) is bridging the gap between the real-setting specimens and the extended space, supporting two way conversations — which allows users to find a species in the application from a real one as well as find a specimen in the real world from a digital species, and forming a closed-loop for the whole exploration flow based on the technology of augmented reality.

All of them support the goal of expanding and uncovering the stories of each displayed specimen and building dialogues between physical installations and digital interfaces.

Visualization Methods

In information visualization workflow, one can often find vastly different ways to visualize the same data set. There are a variety of strategies that drive design decisions about what graph

to choose to encode the data, such as from analytical intent^{7 8 9}, problem-driven¹⁰, from user tasks¹¹, from graphical aesthetics¹², from the effectiveness of visual channels¹³, from data format¹⁴, from the recommendations of BI tools based on the Grammar of Graphics¹⁵ and so on. Thus, designing a visualization is never as easy as using a fixed type of chart to visualize a class of data.



Figure 2: Strategies for building Connected Specimens

For a broader design space, we summarized some concepts in the field of visualization design and human-computer interaction. In Figure 2, the top and bottom rows are common opposing categories, covering data set size, view organization, human-computer interaction paradigm, narrative paradigm, user exploration paradigm, and multi-user supportability. The black ones are the paradigms that have been chosen in our application. We will discuss the first two columns of concepts in this section, as well as the other visualization construction methods involved in this project. The last three sets of concepts will be discussed in the next section.

⁷ Severino Ribeca and Jonathan Schwabish, "The Graphic Continuum", Severinoribeca.One, 2014, <https://www.severinoribeca.one/portfolio-item/the-graphic-continuum/>.

⁸ "Visual Vocabulary", Github, 2016, <https://github.com/ft-interactive/chart-doctor/tree/master/visual-vocabulary>.

⁹ Andrew Abela, "Chart Suggestions—A Thought-Starter", Extremepresentation.typepad.com, 2009, <https://extremepresentation.typepad.com/files/choosing-a-good-chart-09.pdf>.

¹⁰ Tamara Munzner, "A Nested Model For Visualization Design And Validation", *IEEE Transactions On Visualization And Computer Graphics* 15, no. 6 (2009): 921-928, doi:10.1109/tvcg.2009.111.

¹¹ Matthew Brehmer and Tamara Munzner, "A Multi-Level Typology Of Abstract Visualization Tasks", *IEEE Transactions On Visualization And Computer Graphics* 19, no. 12 (2013): 2376-2385, doi:10.1109/tvcg.2013.124.

¹² Geoff Mcghee, "Taking Data Visualization From Eye Candy To Efficiency", National Geographic, 2015, <https://www.nationalgeographic.com/science/article/150922-data-points-visualization-eye-candy-efficiency>.

¹³ William S. Cleveland and Robert McGill, "Graphical Perception: Theory, Experimentation, And Application To The Development Of Graphical Methods", *Journal Of The American Statistical Association* 79, no. 387 (1984): 531-554, doi:10.1080/01621459.1984.10478080.

¹⁴ Yan Holtz and Conor Healy, "From Data To Viz | Find The Graphic You Need", 2018, <https://www.data-to-viz.com/>.

¹⁵ Hadley Wickham, "A Layered Grammar Of Graphics", *Journal Of Computational And Graphical Statistics* 19, no. 1 (2010): 3-28, doi:10.1198/jcgs.2009.07098.

Visualizing Small Data

As mentioned earlier, there are many factors that influence design decisions in the workflow of data visualization. However, it is not only the data structure and the questions asked about the data that determine the outcome of the visualization¹⁶, the size of the data set also affects, and constrains the design strategy of visualizations and human-computer interfaces. Although visualization is always subconsciously associated with big data, such as social network data, urban traffic trajectory data, aviation data, etc., small data also has its unique appeal. Small data studies can answer specific research questions in a more granular way, focusing on specific cases; telling “individual and nuanced stories”; and exploring “in detail and in depth the varied, contextual, rational and irrational ways in which people interact and make sense of the world”¹⁷.

In this project, the data from the IUCN Red List is highly standardized, with a uniform, normalized data format. For each museum or each species, the dataset we need to create is a subset of the global database. That is, we are cutting out individual pieces from a large-scale dataset in a specific context and theme. This echoes Rob Kitchin's view that “all data were, in effect, small data”¹⁸. As a result, the size of the data set in our project is highly predictable — ranging from a few tens to hundreds of data points, because the items are nothing more than a collection of specimens arranged in various forms in different museums or installations, which could not be a large-scale setting. This can well be described as visualizing for small data. Thus, highly standardized data structures and miniaturized dataset sizes offer us the possibility to avoid complex visual rhetoric and create visualizations that are more narrative, storytelling, and individualistic, as well as to customize interaction patterns that are more focused on individuals.

¹⁶ Anne Burdick et al., *Digital Humanities*, n.d.

¹⁷ Rob Kitchin, *The Data Revolution: Big Data, Open Data, Data Infrastructures & Their Consequences* (repr., 27-29: Sage, 2014).

¹⁸ Kitchin, *The Data Revolution*, 27-29.

Unit Visualization

Unit Visualization “describes a class of visualizations where each row of the data-table has a single graphic representation which we refer to as a unit.”¹⁹ Although the concept was only formally introduced in 2015 by Drucker et al. at Microsoft Research, the use of this form in works of known vintage dates back to the 1600s with the invention of Scatterplots, a form of unit visualization based on Geometric shape units, 2D and rectangular coordinate systems, and Map2D layout²⁰. In 2002, a very similar concept — pixel visualization — was proposed by Daniel A. Keim et al. who proposed pixel bar charts that “retain the intuitiveness of traditional bar charts while allowing very large data sets to be visualized in an effective way”²¹. Unit visualization follows on from this work, while both refer to the basic idea of contrasting with aggregate visualizations.

Microsoft Research commercialized this concept with SandDance, a unit visualization constructing system, and this form of visualization has since become commonplace. We can now see it in popular visualizations, such as the 2018 Pudding Cup winner *Twelve years in South Africa's schools*, which uses unit visualization to tell a story of South African devastating education system²².

This is an unparalleled technique that supports visual exploration as well as visual narrative well: it is useful in both complex visual analytics systems and in explanatory visualizations for the general public. Because each data point is represented individually while forming a whole, unit visualization allows users to see the forest and the trees simultaneously, interact

¹⁹ Steven Drucker and Roland Fernandez, "A Unifying Framework For Animated And Interactive Unit Visualizations", Microsoft Research, 2015, <https://www.microsoft.com/en-us/research/publication/a-unifying-framework-for-animated-and-interactive-unit-visualizations/>.

²⁰ Deokgun Park et al., "Atom : A Grammar For Unit Visualizations.", *IEEE Transactions On Visualization And Computer Graphics* 24, no. 13 (2018): 3032-3043, doi:10.1109/TVCG.2017.2785807.

²¹ Daniel A. Keim et al., "Pixel Bar Charts: A Visualization Technique For Very Large Multi-Attribute Data Sets", *Information Visualization* 1, no. 1 (2002): 20-34, doi:10.1057/palgrave.ivs.9500003.

²² Laura Grant and Alastair Otter, "Twelve Years In South Africa's Schools", Passmark.Org.Za, 2018, <http://passmark.org.za/schools/>.

directly with the smallest granularity of data points, naturally support animated transitions, and maintain semantic and attentional constancy²³.

Connected Specimens, in which each digital species item is a unit, exploits the potential of unit visualization to show items as individuals and as a group, facilitate the detail-focused interaction with each item, provide the foundation for animated transitions and enhance the coherency of items in movements among multiple views.

Animated Transition

Animation is a very important cognitive enhancement method in visualization. Animation can be used directly to encode attributes of static data²⁴ and present dynamic streaming data^{25 26 27}, demonstrating complex dynamic changes. Besides, animation reveals insights by supporting visualization transition²⁸, data transition^{29 30 31} and highlighting³². Finally, animation

²³ Drucker, Steven, and Roland Fernandez. "A unifying framework for animated and interactive unit visualizations." Microsoft Research, Aug(2015).

²⁴ Hugo Romat et al., "Animated Edge Textures In Node-Link Diagrams: A Design Space And Initial Evaluation", in *The ACM Conference On Human Factors In Computing Systems*, 2018, 1-13, <https://dl.acm.org/doi/10.1145/3173574.3173761>.

²⁵ Samuel Huron, Romain Vuillemot and Jean-Daniel Fekete, "Visual Sedimentation", *IEEE Transactions On Visualization And Computer Graphics* 19, no. 12 (2013): 2446-2455, doi:10.1109/tvcg.2013.227.

²⁶ Yun Wang et al., "Animated Narrative Visualization For Video Clickstream Data.", in *SIGGRAPH Asia 2016 Symposium On Visualization* (repr., ACM, 2016), <https://doi.org/10.1145/3002151.3002155>.

²⁷ Shixia Liu et al., "Online Visual Analytics Of Text Streams", *IEEE Transactions On Visualization And Computer Graphics* 22, no. 11 (2016): 2451-2466, doi:10.1109/tvcg.2015.2509990.

²⁸ Jeffrey Heer and George Robertson, "Animated Transitions In Statistical Data Graphics", *IEEE Transactions On Visualization And Computer Graphics* 13, no. 6 (2007): 1240-1247, doi:10.1109/tvcg.2007.70539.

²⁹ Pierre Dragicevic et al., "Temporal Distortion For Animated Transitions", in *Proceedings Of The SIGCHI Conference On Human Factors In Computing Systems* (repr., ACM, 2011), <https://doi.org/10.1145/1978942.1979233>.

³⁰ Fan Du et al., "Trajectory Bundling For Animated Transitions", in *The ACM Conference On Human Factors In Computing Systems*, 2015, 289-298, <https://doi.org/10.1145/2702123.2702476>.

³¹ Yong Wang et al., "A Vector Field Design Approach To Animated Transitions", *IEEE Transactions On Visualization And Computer Graphics* 24, no. 9 (2018): 2487-2500, doi:10.1109/tvcg.2017.2750689.

³² Manuela Waldner et al., "Attractive Flicker — Guiding Attention In Dynamic Narrative Visualizations", *IEEE Transactions On Visualization And Computer Graphics* 20, no. 12 (2014): 2456-2465, doi:10.1109/tvcg.2014.2346352.

is a general tactic to evoke audience emotions³³, connect people and data³⁴, and improve engagement³⁵.

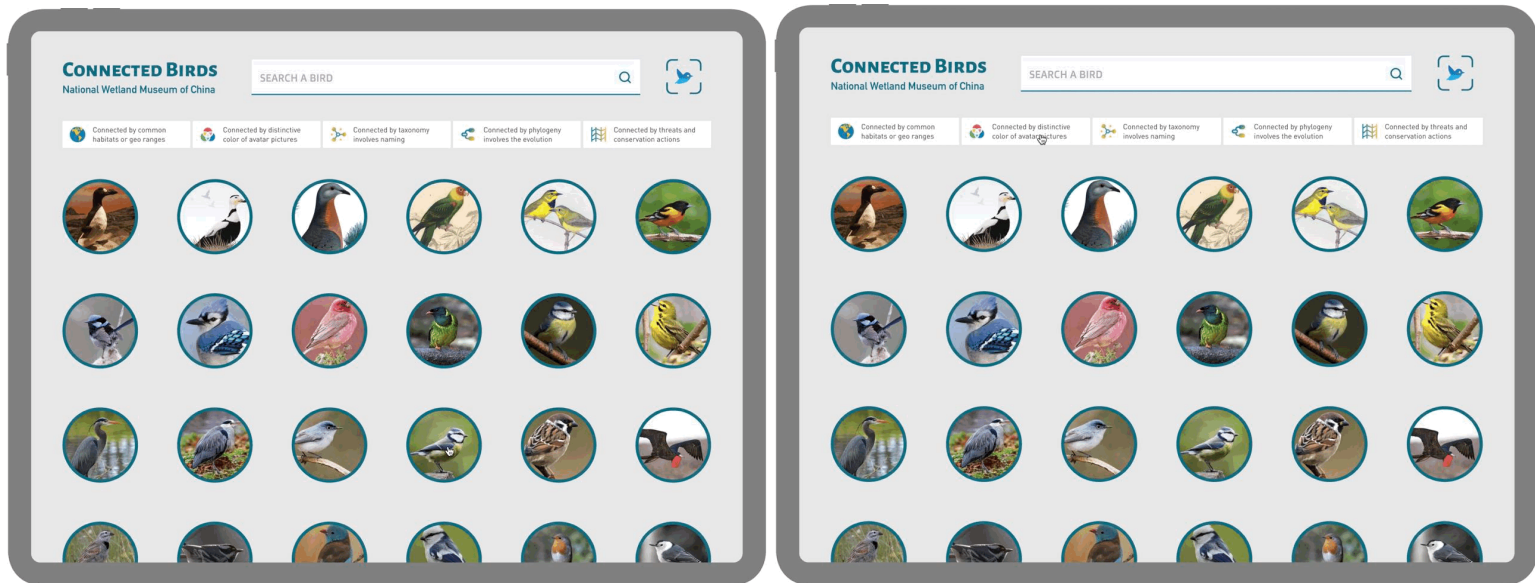


Figure 3: Transitions as the connective threads between individual maps

In Connected Specimens, we use the animation technique (Figure 3) to create synergies with the unit visualization technique mentioned above. Our primary purpose in using animations is to drive transitions, bridge layout breaks between views, and amplify the user's perception of the data. That is, when the user switches views, the data points themselves remain the same, but are repositioned according to the view's theme and the visualization format is reorganized to a new layout. We need transitions between locations to help the user maintain continuity of the mental model. Also, during and after the transition, the previously selected object will remain highlighted and selected, which will not only help the user to follow the target, but also to quickly locate its new position.

³³ Fengjiao Peng et al., "A Trip To The Moon: Personalized Animated Movies For Self-Reflection", in *Proceedings Of The 2018 CHI Conference On Human Factors In Computing Systems* (repr., ACM, 2021), 1-10, <https://doi.org/10.1145/3173574.3173827>.

³⁴ Fred Hohman et al., "Communicating With Interactive Articles", *Distill* 5, no. 9 (2020), doi:10.23915/distill.00028.

³⁵ Fereshteh Amini et al., "Hooked On Data Videos: Assessing The Effect Of Animation And Pictographs On Viewer Engagement", in *Proceedings Of The 2018 International Conference On Advanced Visual Interfaces*, 2018, 21.

No Multiple Coordinated Views

Multiple coordinated views are widely used in building visual analytics tools, and more and more business intelligence tools have added support for this technique³⁶ in recent years.

This technique is considered to be the beauty and power of visualization, because visualization is more than just a visual and rhetorical representation of established knowledge, descriptive statistics; another important mission of visualization is to support data exploration and discovery of unknown knowledge^{37 38} by harnessing the intelligence of the human brain and visual perception as the non-linear channel with the greatest bandwidth of the human senses. Multiple coordinated views support exactly this kind of data exploration and knowledge discovery: multiple views present different aspects of the data in the same page, based on the same data fields, establishing an interrelated coordination³⁹ and forming a highly coupled human-computer interactive system, making it possible for humans to explore potential patterns.

While not yet common in narrative visualization and data journalism for the general public, the technique has been used by many non-scientific business design teams and individuals, for example in Clever Franke's transport system planning tool⁴⁰ for Abu Dhabi's Department of Municipalities and Transportation and Christian Laesser's train capacity prediction tool⁴¹ for Deutsche Bahn AG. During the COVID-19 epidemic, Tencent in China and the Visualization and Visual Analytics Lab at Peking University experimented with this technology on mobile devices and in mass communication, and the author had the privilege to lead this design.

³⁶ Ross Perez, "Use Parameters To Add Multiple Views To Your Viz", Tableau, 2015, <https://www.tableau.com/about/blog/2015/7/lrtt-parameters-alter-your-viz-40767>.

³⁷ J.J. Thomas and K.A. Cook, "A Visual Analytics Agenda", *IEEE Computer Graphics And Applications* 26, no. 1 (2006): 10-13, doi:10.1109/mcg.2006.5.

³⁸ Jonathan C. Roberts, "State Of The Art: Coordinated & Multiple Views In Exploratory Visualization", in *Fifth International Conference On Coordinated And Multiple Views In Exploratory Visualization (CMV 2007)*, 2007, 61-71.

³⁹ Paulo Igor Alves Godinho et al., "PRISMA - A Multidimensional Information Visualization Tool Using Multiple Coordinated Views", in *2007 11Th International Conference Information Visualization (IV '07)* (repr., IEEE, 2007), 23-32.

⁴⁰ CLEVER°FRANKE, "Abu Dhabi Birdseye", 2020, <https://www.cleverfranke.com/work/birdseye>.

⁴¹ Christian Laesser, "Peak Spotting", 2017, <https://www.christianlaesser.com/projects/peak-spotting-train-data-visualization.html>.

However, in some scenarios, this technique needs to be used with caution, even though “one of the exciting aspects of visualization is the ability to explore the connections between different datasets”⁴². In museum scenarios, the short time visitors spend in front of an exhibit and the lack of prior knowledge of the data pose challenges to the design of a visual exploration system⁴³. An empirical study conducted by Joyce Ma et al. found that “adding a representation of a secondary data variable seemed to increase the time to first interpretation” and provided the following two related insights:

1. control the amount of information presented on each page, retaining the most important information only.
2. additional representations need to be carefully considered.

This insight and philosophy was critically applied to the design of our system. For most of the views, we kept the single-view design and made sure that the form of the view was closely related to the theme.

At the same time, however, we believe that this principle is not quite absolute. There are too many variables that affect the user's interpretation of the data, which means that we have many other options to hedge against the understanding challenges that multiple views pose to non-expert users.

⁴² Joyce Ma, Kwan-Liu Ma and Jennifer Frazier, "Decoding A Complex Visualization In A Science Museum - An Empirical Study", *IEEE Transactions On Visualization And Computer Graphics*, 2019, 1-1, doi:10.1109/tvcg.2019.2934401.

⁴³ Ma, Decoding a Complex Visualization.

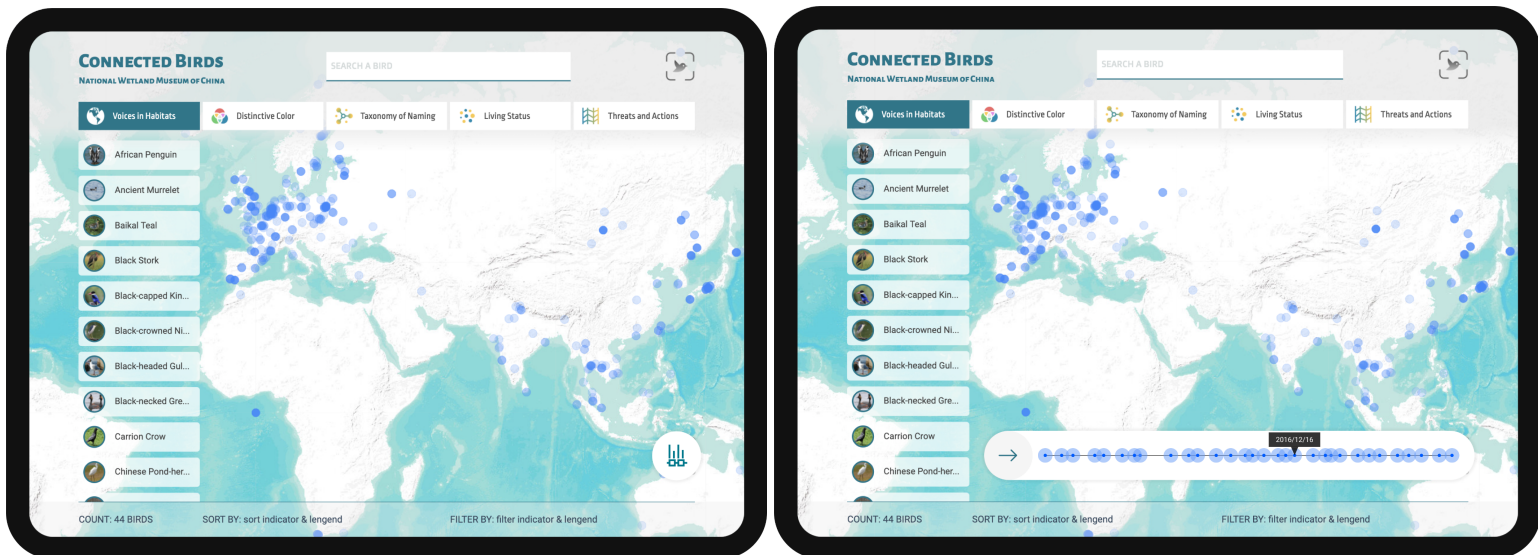


Figure 4: A pilot design of an unfolding strategy for the simplified multiple coordinated views for the geographic view. This has not been implemented in the published edition.

We tried to embed a timeline in the geographic view (Figure 4) through a pilot design, with the aim of supporting a dual projection of data points in both geographic and temporal coordinate systems. This design maximizes the value of unlocking the data: not only spatially observing the distribution of sound recordings of one avian species, but also perceiving their flow and evolution in temporal — yearly and daily — dimensions to reveal the potential trend⁴⁴, and supporting interactive queries from time to space or from space to time. To soften the challenges of multiple coordinated views, we have thought through several strategies from a user-experience design perspective.

1. Simplifying auxiliary information, including axis scales, text annotations, etc. Limitedly increase the amount of information that users need to decode to control the cognitive load. Those auxiliary information, which is hidden by default, can be displayed by means of interactions, such as clicks or hovers.
2. By default, only a single view (map) is displayed, and the timeline is kept collapsed. This ensures that the user only decodes one dimension of information when first entering the page.

⁴⁴ Katy Börner and David E Polley, *Visual Insights: A Practical Guide To Making Sense Of Data*, 3rd ed. (repr., MIT Press, 2014).

3. Giving a semantic graphical hint on the button to guide the user's clicking behavior of expanding the timeline.

Thus, with the above design, the user goes through a process from single view to multiple (only two here) coordinated views, and this process is actively triggered by the user through interaction, which means that the user has probably already finished decoding the view being viewed and has developed a desire to explore it further through other dimensions. This stepwise method of unfolding views not only provides an initial context for the user, but also helps them build up a mental expectation of the new view before clicking through the hint, avoiding them being overwhelmed by the many unfamiliar views. But certainly, the effectiveness of this design needs to be further evaluated through user experiments.

ENGAGEMENT MODES FOR VISITORS

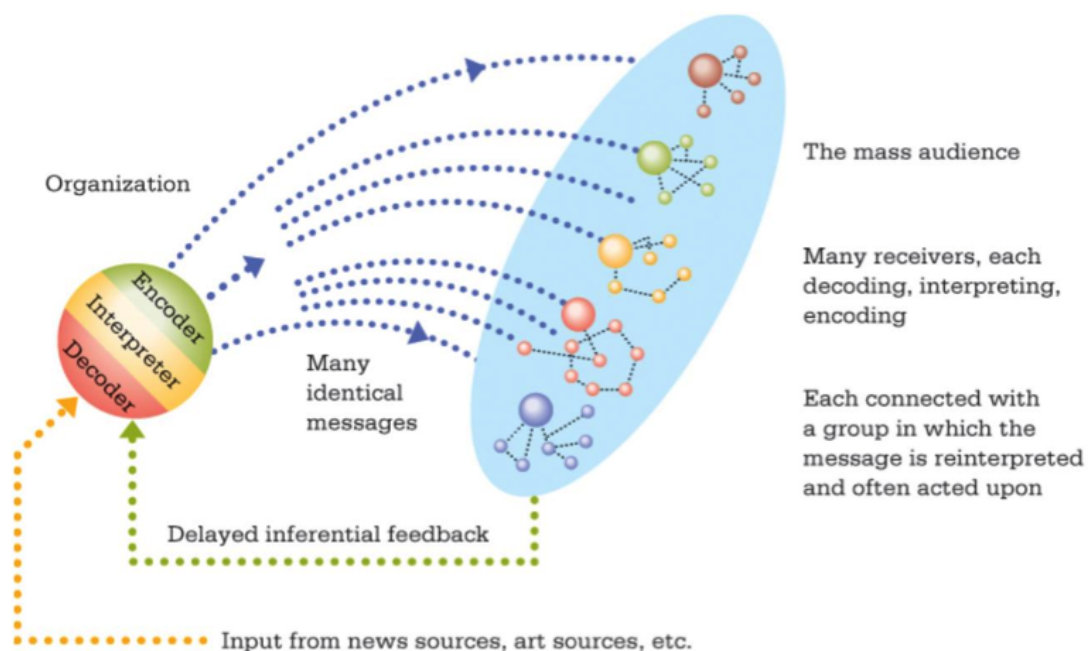


Figure 5: Schramm's model of mass communication⁴⁵

Creating superior engagement modes for a tool is to make users enjoy using it, benefit from it, and become co-creators to shape it. The audience in exploring Connected Specimens is beyond a passive receiver, decoder, and reader. The interaction process here is similar to

⁴⁵ Wilbur Schramm and Donald F Roberts, *The Process and Effects of Mass Communication* (repr., Urbana, IL: University of Illinois Press, 1977).

*Schramm's model of mass communication*⁴⁶. The center of this model is the content authors who offer “many identical messages” and the audience consists of individual users who decode and interpret the “messages” respectively. In Connected Specimens, users are empowered to choose the visualization views along their interests — which may develop during the exploration process — and interact with the data points by myriad data manipulation options, which ultimately supports them to curate their own experiences and form disparate user flows.

This chapter we will discuss some cognitive facilitation strategies in engagement modes in data-driven exploration and storytelling and designing for museums.

Understanding the Users

According to an informal report⁴⁷ from Smithsonian Institution, some of the main purposes for visitors to go to a museum are:

1. provide educational or enriching experiences for their children
2. learn something new about a particular topic or a variety of topics
3. have a social experience with friends
4. relax, recharge, and have a fun time

The report also mentioned that museums are obsessed with social interaction because it is a key indicator of learning — it would be easy for a person to retain what they have learned in long-term memory. In addition, encouraging visitors to be participants from listeners, providing multiple perspectives, introducing immersive experiences would be some strategies to uncover the “digital superpowers”⁴⁸, according to this report. Being more specific for cultural heritage data, Eva Mayr et al. define these kinds of users as casual users: “users in everyday, non-work contexts”⁴⁹, and summarized their exploration behaviour

⁴⁶ Schramm and Roberts, *Mass Communication*.

⁴⁷ The Information Session of Activating Smithsonian Open Access Challenge, 2020.

⁴⁸ The Information Session, 2020.

⁴⁹ Eva Mayr et al., “Visualization Of Cultural Heritage Data For Casual Users”, in *IEEE VIS Workshop On Visualization For The Digital Humanities* (repr., IEEE, 2016), http://publik.tuwien.ac.at/files/PubDat_250950.pdf.

that comes “without concrete information needs, they rather search for something interesting and engaging”⁵⁰. These insights provided us with a very valuable reference for designing museum experiences.

Designing for Data Exploration

User-directed Exploration

According to Eva Mayr et al., given the fact that “no predefined user tasks exist; rather users set their own goals and freely explore the collection,”⁵¹ design requirements include “providing conceptual orientation”, “supporting exploration search”, providing entertaining learning experience, lowering cognitive load, and enabling intuitive interactions.

Based on these five principles and the use scenarios of Connected Specimens, we conceptualized several design principles, and applied some of them to the design practice.

1. Providing guidance

Guidance refers to splash pages, navigations, annotations, commentary, background knowledge and any other semantic information to provide context for users’ better comprehension and support them for directing their ways through the visiting.

2. Enhancing the self-explanatory and intuitive features

Providing hints and signifiers for user interactions; using natural metaphors⁵² and naïve analogy⁵³ philosophy for casual users.

3. Being pan-entertainment

Designing for museum experiences should be relaxation-oriented, because most of the users would not pursue a serious educational text. Educational goals in this environment are not as specific as a class; they rather are often presented in a fun and educational way. To achieve this style, some strategies could be incorporated:

⁵⁰ Mayr, et al. Visualization of Cultural Heritage Data.

⁵¹ Mayr, et al. Visualization of Cultural Heritage Data.

⁵² Daniel Keim et al., *Mastering The Information Age – Solving Problems With Visual Analytics* (repr., Goslar: Eurographics Association, 2010), 109-130.

⁵³ Douglas R Hofstadter and Emmanuel Sander, *Surfaces And Essences: Analogy As The Fuel And Fire Of Thinking* (repr., New York: Basic Books, 2013).

keeping things simple to lower the cognitive load; gamification; being informative as well as inviting⁵⁴.

Multi-user Exploration

Please refer to [Reality-based Interaction](#) in the Chapter of Designing for Learning and Cognition.

Designing for Data-driven Storytelling

Storytelling can be valuable to visualization^{55 56}, and visualization also can have contribution to storytelling^{57 58 59}. When storytelling meets visualization, a variety of narrative visualizations^{60 61} are created, resulting in data-driven visualization works — across myriad thematic and cultural contexts — that are known to countless ordinary people and becoming probably the most popular form of visualization known to the public.

⁵⁴ Marian Dörk, Sheelagh Carpendale and Carey Williamson, "The Information Flaneur: A Fresh Look At Information Seeking", in *Proceedings Of The SIGCHI Conference On Human Factors In Computing Systems* (repr., ACM, 2011).

⁵⁵ Gershon, Nahum, and Ward Page. "What storytelling can do for information visualization." *Communications of the ACM* 44.8 (2001): 31-37.

⁵⁶ Kosara, Robert, and Jock Mackinlay. "Storytelling: The next step for visualization." *Computer* 46.5 (2013): 44-50.

⁵⁷ Lee, Bongshin, et al. "More than telling a story: Transforming data into visually shared stories." *IEEE computer graphics and applications* 35.5 (2015): 84-90.

⁵⁸ Ma, Kwan-Liu, I. Liao, J. Frazier, H. Hauser, and H.-N Kostis. 2012. "Scientific Storytelling Using Visualization." *IEEE Computer Graphics and Applications* 32 (1): 12–19.

⁵⁹ Lankow, Jason, Josh Ritchie, and Ross Crooks. 2013. *Infographics: The Power of Visual Storytelling*.

⁶⁰ Segel, Edward, and Jeffrey Heer. "Narrative visualization: Telling stories with data." *IEEE transactions on visualization and computer graphics* 16.6 (2010): 1139-1148.

⁶¹ Stolper, Charles D., et al. "Emerging and recurring data-driven storytelling techniques: Analysis of a curated collection of recent stories." *Microsoft Research*, April 3 (2016): 2016.

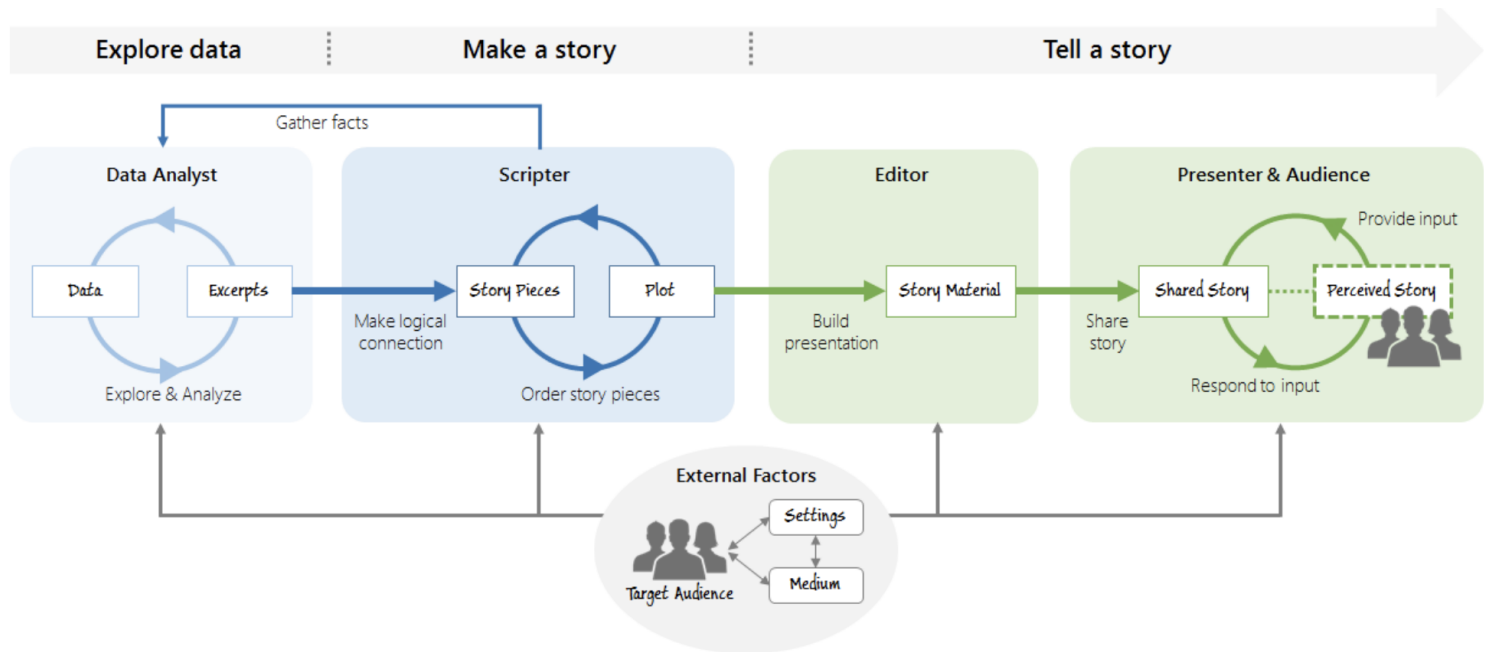


Figure 6: The storytelling process proposed by Lee et al.⁶²

Lee et al. summarized a workflow for data-driven storytelling (Figure 6), delineating three stages from exploring data, to make a story, and to tell the story. Each stage involves different roles: from data analyst, to scripter, editor, presenter and user.

Author-directed Storytelling

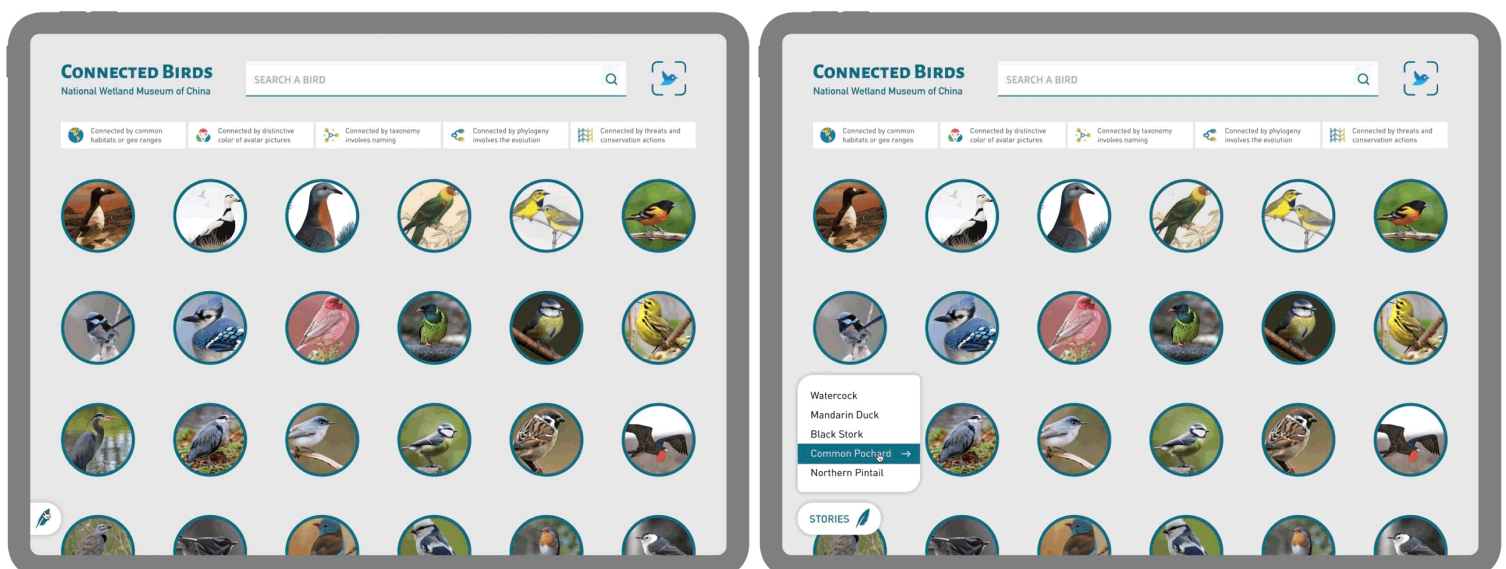


Figure 7: a user guidance design that leverages author-directed storytelling

⁶² Lee, et al. More than telling a story.

The huge exploration space of this application provides users with the freedom to explore but also poses challenges: users can feel lost and overwhelmed by the sheer number of options to interact with. Fortunately, the huge exploration space of this application provides a natural ground for, or underlies the possibility of, building a wide variety of data-driven narrative flows. We therefore consider introducing pre-set stories of selected avian species (Figure 7) as the guidance to help users in turn better understand the characteristics and usage of each view. Figure 9 shows a pre-set story of a given bird, which provides more sufficient context and leads users to explore the application with a better sense of direction and purpose. Stories could be pre-prepared by the operators or organizers of a museum. They could “make logical connections”, “order story pieces”, “build presentation” and then the audience could “perceive stories”⁶³. Museums can even offer visitors different digital experiences by season and thematic exhibitions.

It is not an uncommon practice to introduce storytelling into an exploration system. The introduction of supplemental author-directed storytelling here not only does not limit the discovery properties of the tool, but opens up a plethora of possibilities for exploration: educate and guides the user, thus reducing the cognitive challenge of having too much space to explore; allow the user to “document the discovery process” to “justify and reproduce insights”⁶⁴; and increase the user's understanding of the availability of different views by introducing gamification⁶⁵ and other means.

User-directed Storytelling

As mentioned above, in addition to the preset species from authors, it is possible to open the system to users to create data stories that are personalized, record and save insights, and share knowledge. Moreover, Connected Specimens as an interactive system compensates for the shortcomings of using images and videos for process documentation: “lack of full

⁶³ Lee, et al. More than telling a story.

⁶⁴ Samuel Gratzl et al., "From Visual Exploration To Storytelling And Back Again", *Computer Graphics Forum* 35, no. 3 (2016): 491-500, doi:10.1111/cgf.12925.

⁶⁵ Fengfeng Ke, Kui Xie and Ying Xie, "Game-Based Learning Engagement: A Theory- And Data-Driven Exploration", *British Journal Of Educational Technology* 47, no. 6 (2015): 1183-1201, doi:10.1111/bjet.12314.

provenance information and context”, difficulty in editing, and contradiction with non-linear exploration processes⁶⁶. This not only resonates with the concept of “users become co-creators to shape it” in Schramm's mass communication model mentioned above, but also echoes the process of user feedback in Lee's model.

Designing for Learning and Cognition

Reality-based Interaction

Relationship Model	Pure Digital	Digital-To-Physical	Physical-To-Digital	Physical-Digital-Physical
Content Source	Author-directed Content			
	User-generated Content			
Interaction Mode	Single-user Interaction			
	Multi-user Interaction			
		Pervasive Computing & Handheld Interaction		
		Social Interaction		
Interaction Paradigm	WIMP-Based Interaction	Reality-Based Interaction (RBI)		
Socialization Level	Screen Awareness	Environment Awareness		
		Social Awareness		

Figure 8: the five potential modes of user engagement from the projects related to museum informatics I summarized.⁶⁷ WIMP stands for “windows, icons, menus, pointer”.

We now have a myriad of digital experiences created for museums. They deserve recognition for their efforts to make museum resources accessible, creating virtual tours for many people who do not have access to museums, such as NMNH's 3D Virtual Tour⁶⁸. But many of these works are often purely digital, simply recreating existing library collections in cyberspace, and there is no dialogue with the museum's real collections. For those who do visit the museum, their experience is not much improved from what it used to be.

⁶⁶ Gratzl, et al. From Visual Exploration to Storytelling.

⁶⁷ Robert J.K. Jacob et al., "Reality-Based Interaction: A Framework For Post-WIMP Interfaces", in *In Proceedings Of The SIGCHI Conference On Human Factors In Computing Systems* (repr., ACM, 2008), 201-210.

⁶⁸ Virtual Tour | Smithsonian National Museum of Natural History.
<https://naturalhistory.si.edu/visit/virtual-tour>

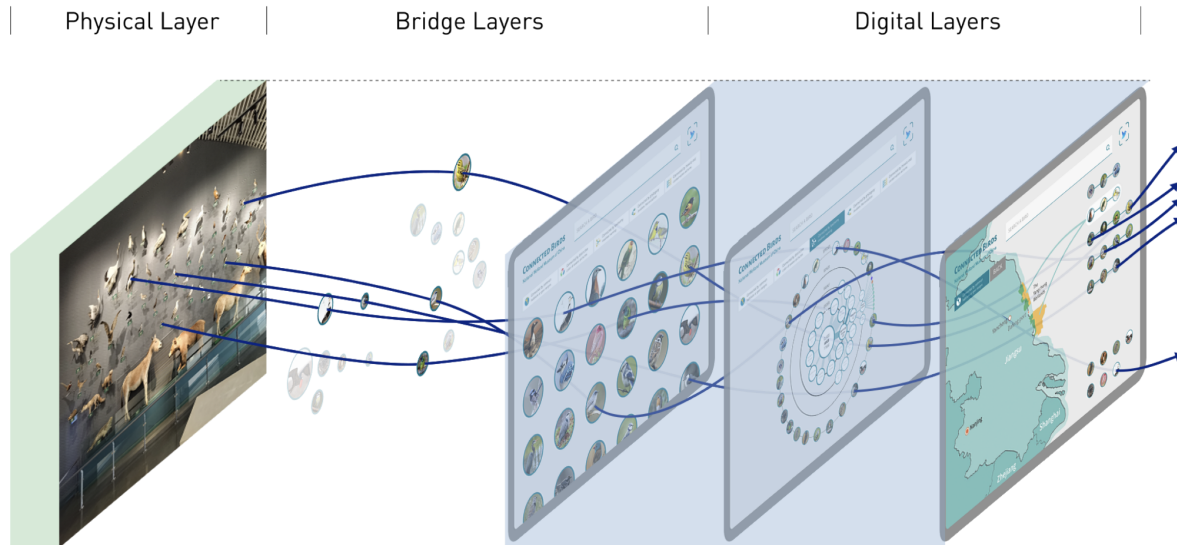


Figure 9: the three layers of the exploratory objects. Our modeling of layers suggests both technical and rhetorical structures — from the layered “technical stack” that makes such data infrastructures possible, to the layers of information that visitors can “dig into” to satisfy their curiosity.

To improve the experience for the in-person experiences in a museum, we introduced a physical-digital-physical model (Figure 8&9) to support two-way conversations. That is, users could uncover the stories of the displayed specimens by scanning the real one, and in turn locate a real specimen in the museum from a digital one. This mode uses the screen as a bridge or information hub, and provides the opportunity for users to move the computing environment out of the screen and transitioning from a WIMP-based user interface to the reality-based interaction⁶⁹ anytime they want.

⁶⁹ Jacob et al., "Reality-Based Interaction," 201-210.

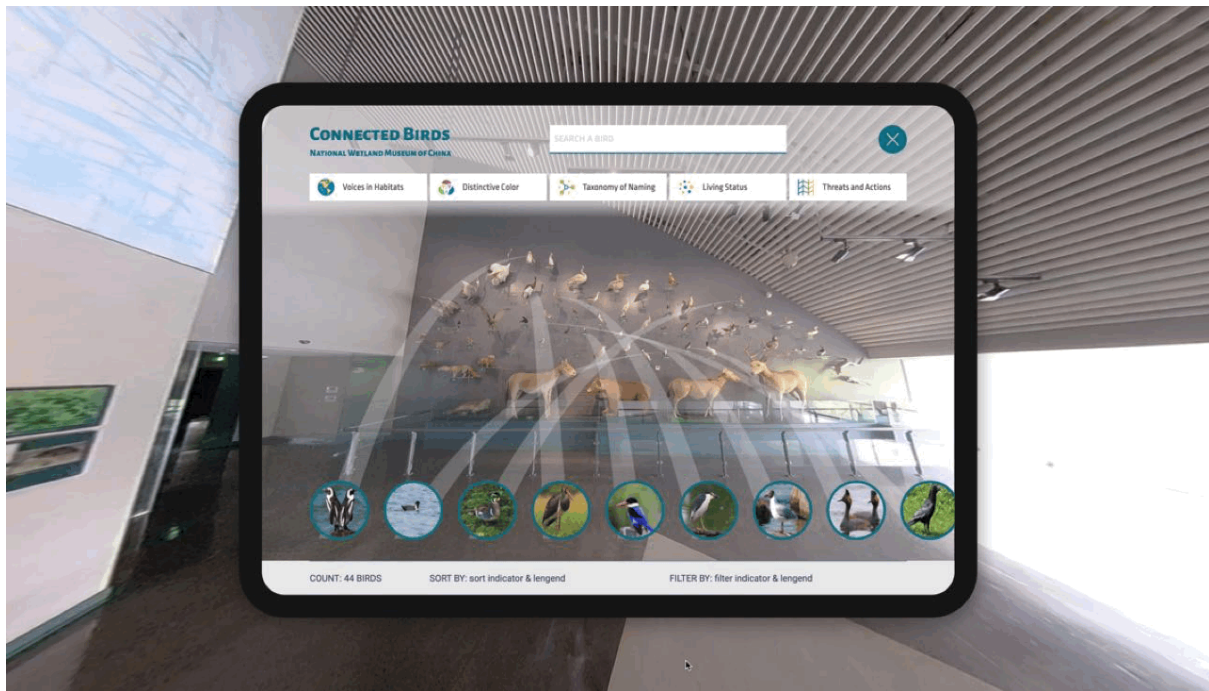


Figure 10: an imitation of the recognition functionality in the real environment.

This firstly encourages users to focus more on human-to-environment and human-to-human communication than human-computer-interaction. User activities are extended from an isolated experience to a social experience.

Moreover, the recognition mode enhances the learning process for visitors. Whenever they lose the real-setting target when they are viewing the items on the device, or whenever they change a selected item but are curious about which one it is in the real setting, they could easily reconnect the world around them to relocate the specimens. They do not need to recall an object any more but recognize it instead. This largely saves their memory and cognition resources, benefit the effective delivery of the information⁷⁰, thus amplifying the effectiveness of the visualization and users' understanding.

Another scenario would be to construct a collective experience by enabling multiple users to explore the specimens at the same time. In the multi-user exploring mode, two users could

⁷⁰ Michelle A. Borkin et al., "Beyond Memorability: Visualization Recognition And Recall", *IEEE Transactions On Visualization And Computer Graphics* 22, no. 1 (2016): 519-528, doi:10.1109/tvcg.2015.2467732.

scan different avian specimens respectively, and compare them in the device. This would not only enhance the data exploration process, but also the human-to-human communication.

In short, a good combination of engagement modes should not be limited to the relationship between people and screens, but should ultimately benefit the relationship between people and environments, and people and people.

WORKFLOWS FOR MUSEUMS AND INSTITUTIONS

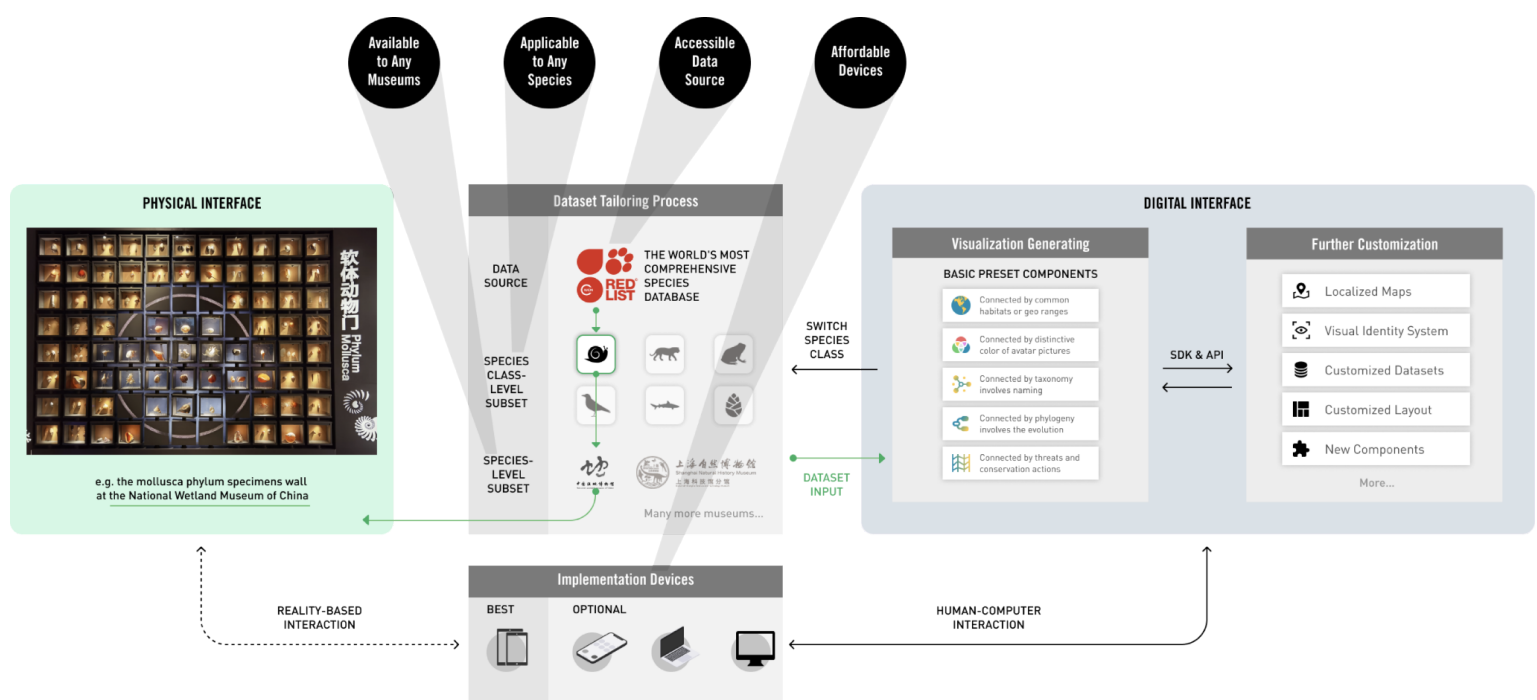


Figure 11: designing an available, applicable, accessible, and affordable museum experience.

Designing for Affordability

In addition to the "field experience in museums that needs to be improved" mentioned in the previous section, the museum experience in less developed regions seems to be forgotten.

In my interview with Anna Gedal, who has been creating digital experiences for museums for around 5 years, she said "Few institutions create them on their own, they normally hire museum consulting firms with specialized skills to build them. The costs are high. Even a basic touch screen kiosk can cost a lot, so many institutions cannot afford it. More broadly, most institutions do not have the tech infrastructure to support tech."

Those museums also have rich collections of flora and fauna, but they do not have the funds either to implement expensive and intensive infrastructure upgrades or to hire professional agencies to customize fancy digital experiences for them. There is an invisible divide between museums in developed and underdeveloped regions.

The ethical commitments underlying this work is the fact that we want to develop tools that pool the resources of museums so all can benefit, especially less-well-resourced institutions.

This application does not require head-mounted augmented reality glasses and does not require an expensive multi-touch table, although it could have the ability to support those devices. The basic computation environments required are tablets or even mobile phones. This could achieve a balance with affordability and the integrity of the digital experience.

Designing for Accessibility

The concept of accessibility already includes affordability in a way. Affordability can be thought of as financial accessibility. But in a more general sense, Specimens Specimens is also highly accessible. While it is originally designed to enhance the interactivity and narrative of offline curatorial resources, the application itself can stand on its own as well by disabling the recognition module, which would be exactly the same as other pure digital online museum experiences.



Figure 12: flashcards and posters of avian species, source: [Bird Nature Study Printables](#)

What's even more exciting is that this offline experience can be extended to a wider range of spaces: for example, museums can make tailored flashcards and posters for kids, so that children can be inspired and informed at home by scanning photographic specimens.

Designing for Universality

The normative and organized nature of the IUCN Red List data makes it possible for our work to support not only a broad range of museums, but also universal species. By simply providing the scientific name of the species, a small dataset specific to that species could be customized. This system was designed not just for avian species, but for all species. More interestingly, it is possible that there are shared or specific views for each type of species. For example, for birds, there may be a view dedicated to exploring the differences in flight altitude of different birds, while for plants, there may be a view for exploring their preferences for temperature and humidity; but a view like the taxonomy can be shared by all types of species.

Designing for Extensibility

While the application makes a lot of adaptation efforts for democratization, it also retains the flexibility to be extensible. As we can see in the "Digital Interface" in Figure 13, in addition to the basic preset modules, we will also provide APIs⁷¹ and SDKs⁷² in the future for further development needs of museums, including support for localized maps, custom museum elements, custom enhanced datasets, custom exploratory modules, and more.

⁷¹ application programming interface

⁷² software development kit

CONCLUSIONS

Connected Specimens provides a framework for museums to enhance the visitors' specimen-viewing experiences with a low cost and a high applicability. This framework expands the stories of the collections in (1) enhancing the ways metadata can encode individual specimens and their collections (2) and visualizations can allow visitors access to that additional contextualizing information; (3) expanding the scope of institutions who can take advantage of such tools without requiring expensive infrastructure upgrades (4) and general users who can flexibly access this experience temporally and spatially.

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